

AI-Powered Workflow System

Built to Scale with Embedded Risk Controls

1 Business Need

- Ad hoc GenAI work left conversations, prompts, sources, and outputs scattered across tools and sessions.
- Hallucinations, retrieval failures, and contradictions could go unnoticed and spread to related outputs.
- Standards, decisions, and approved outputs were not consistently carried into later work, causing rework and drift.

2 Design Goal

- Build one repeatable workflow that can support more workstreams and output types.
- Set clear intended-use boundaries, review every stage, recheck affected work after changes, and require human approval for final release.

3 Solution Architecture

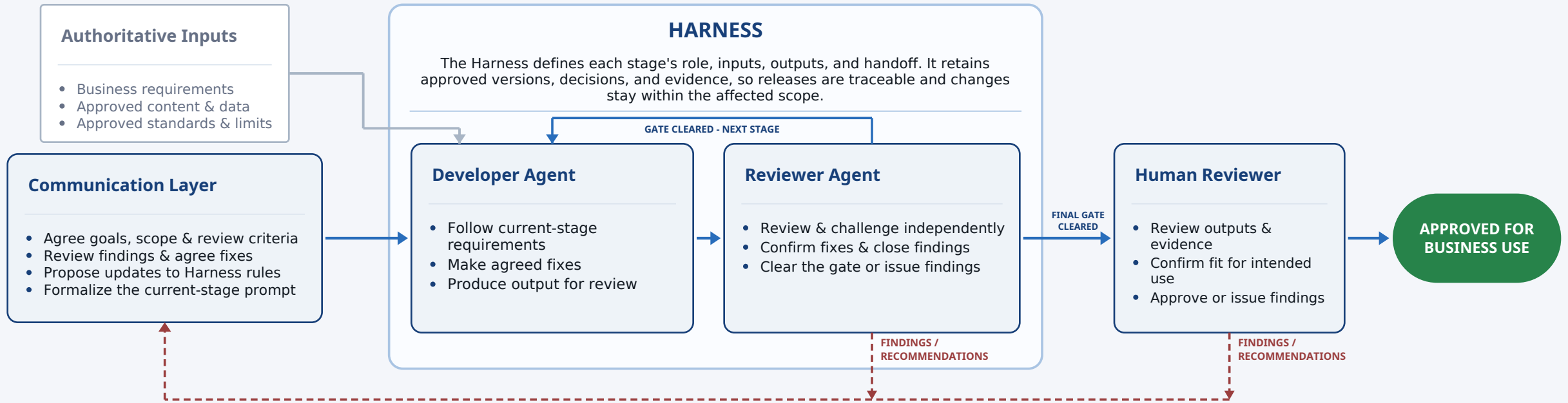
- Communication Layer: Turns agreed goals and findings into clear prompts.
- Developer Agent: Completes each stage and makes agreed fixes.
- Reviewer Agent: Independently reviews and challenges each stage output.
- Harness: Runs the stage sequence, applies review gates, and records state and evidence.
- Human Reviewer: Approves final release.

4 Results

- Repeating Production: Produces consistent outputs from business requirements and standards.
- Current Use: Supports four college courses and two separately delivered class sections, with expansion underway.
- Controlled Releases: Keeps related outputs aligned, approved, and traceable.

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Agent-Harness Architecture: Scalable Execution with Embedded Risk Controls



Scalability

- Shared skills and standards capture agreed methods and improvements. New workstreams reuse them with their own requirements.
- The Harness breaks complex work into stages, coordinating production and review across related outputs.
- Saved progress and approved outputs let work continue without starting over. Updates focus on the affected work.

Risk Controls

- Human-AI discussion uses approved sources to define each stage's goal, scope, and review criteria, then formalizes the prompt.
- Developer Agent follows the current-stage prompt, using approved inputs and making only agreed changes.
- Reviewer Agent checks outputs in a separate context. Findings return to the Communication Layer for a revised prompt, then the work is corrected and resubmitted.
- Harness separates workstreams, passes only cleared work forward, and keeps approved versions and review evidence. Changes are revalidated before use; final release needs human approval.